RESEARCH PAPER

System Capacity & Care Load Analytics for Unaccompanied Children

An Excel-based exploratory analysis of reported custody, care, transfer and discharge data

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Prepared for academic / internship submission. Findings are descriptive and limited to the supplied aggregate dataset.

Abstract

This paper presents an exploratory analysis of reported Unaccompanied Alien Children (UAC) custody and care counts, using the project workbook supplied for the internship. The cleaned dataset contains **720 dated observations** between **12 January 2023** and **21 December 2025**. The observed combined load declined from **6,619** at the first reporting point to **2,502** at the latest point, while the maximum observed combined load was **11,762** on 20 December 2023. Aggregate reported HHS discharges exceeded CBP-to-HHS transfers over the observed records. The paper emphasizes that missing reporting dates, the absence of actual capacity and staffing data, and the aggregate nature of the measures prevent conclusions about verified over-capacity or individual-level outcomes.

Keywords: Unaccompanied children; care load; custody; capacity planning; transfer flows; exploratory data analysis; data quality.

Contents

1. Introduction and background • 2. Problem statement and objectives • 3. Dataset and scope • 4. Data preparation and methodology • 5. Key findings • 6. Time-based analysis • 7. Capacity and operational interpretation • 8. Recommendations • 9. Limitations • 10. Conclusion • Appendix A. KPI definitions • Appendix B. Data quality summary

1. Introduction and background

The UAC care pathway includes children recorded in U.S. Customs and Border Protection (CBP) custody and children in Department of Health and Human Services (HHS) care. These are distinct stages and reporting measures. A consolidated view of the reported counts and related transfers and discharges can help analysts monitor observed load patterns, understand flow balance, and frame questions for operational planning.

This project develops a repeatable workbook-based analytical view rather than a predictive or causal model. It organizes the supplied data, derives consistent indicators, and documents how those indicators should be interpreted.

2. Problem statement and objectives

The source measures are distributed across multiple fields. Without a structured analysis, it is difficult to compare CBP custody and HHS care snapshots, identify high observed combined-load periods, assess reported transfer/discharge balance, and communicate changes consistently.

Primary objectives

• Quantify reported CBP custody, HHS care, and combined load.
• Examine changes between observed reporting points and over time.
• Compare transfers out of CBP custody with HHS discharges.
• Build monitoring indicators that may inform capacity-planning discussions.

Secondary objectives

• Validate and document data quality.
• Provide a repeatable reporting framework and practical recommendations.

3. Dataset and scope

The supplied source contained 1,170 spreadsheet rows, including 450 fully blank rows. After removing fully blank rows from the analytical copy and retaining rows with parseable dates and populated source values, **720 dated observations** remained. The observed date range is 12 January 2023 to 21 December 2025. This interval spans 1,075 calendar days, of which 355 dates have no source record. Missing dates are retained as missing observations and are not converted to zero.

The analysis uses aggregate counts. It does not contain person-level records, shelter-level capacity, occupancy thresholds, staffing, length of stay, waiting time, or placement-time information.

4. Data preparation and methodology

The project workbook's cleaned-data and analysis sheets were used as the analytical basis. The preparation process removes completely blank rows from the analysis copy, parses and sorts dates, converts count fields to numeric values, and checks duplicate dates, missing numeric values, and negative counts. The original raw-data sheet is retained in the workbook for traceability.

Derived measures

• **Total System Load** = reported children in CBP custody + reported children in HHS care.
• **Net Intake Pressure** = children transferred out of CBP custody – children discharged from HHS care. This is a flow-balance proxy, not a complete system-wide stock-flow equation.
• **Endpoint change** = (latest observed total load ÷ first observed total load – 1) × 100.
• **Discharge offset ratio** = aggregate HHS discharges ÷ aggregate transfers out of CBP custody × 100.
• Rolling indicators use windows of 7 or 14 *observed records*, not necessarily consecutive calendar days.

5. Key findings

Measure	Result
Dated observations	720
First observed combined load	6,619 (12 Jan 2023)

Measure	Result
Latest observed combined load	2,502 (21 Dec 2025)
Peak observed combined load	11,762 (20 Dec 2023)
Peak HHS care snapshot	11,516 (20 Dec 2023)
Peak CBP custody snapshot	531 (04 Feb 2024)
Endpoint change	-62.2%
Aggregate transfers	92,641
Aggregate HHS discharges	124,853
Aggregate discharge offset ratio	134.8%
Aggregate net intake pressure	-32,212

The first-to-latest endpoint comparison is a -62.2% change across the observed endpoints; because the series has reporting gaps, it should not be interpreted as a continuous daily trend or attributed to a particular cause.

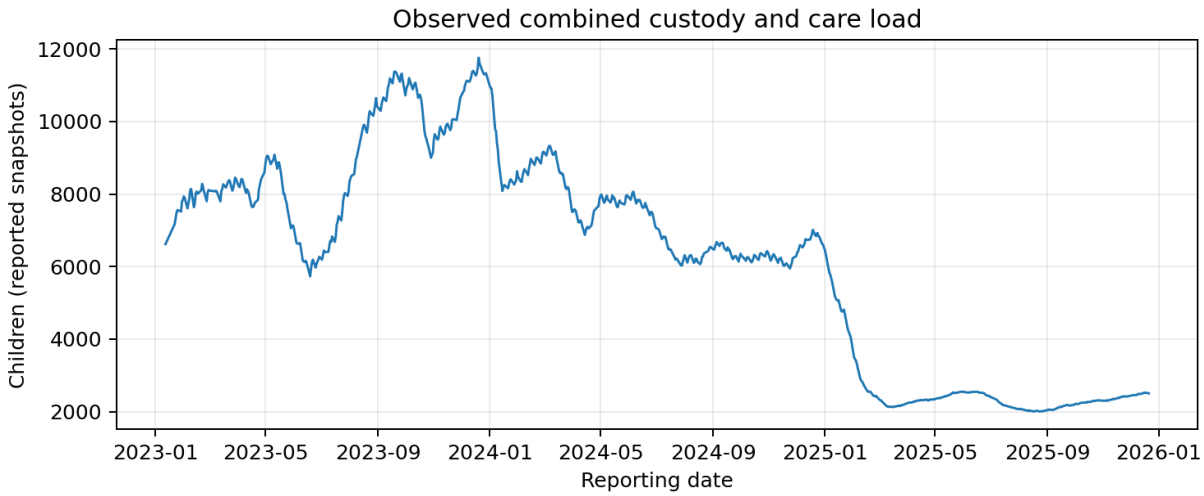


Figure 1. Combined reported CBP custody and HHS care snapshots at available reporting dates. Gaps between observations are not filled or interpolated.

6. Time-based analysis

The annual summary in the workbook shows materially different average observed combined loads across the three calendar years. The annual means are calculated from available records only; reporting coverage is incomplete and uneven, so these values are descriptive summaries rather than complete daily averages.

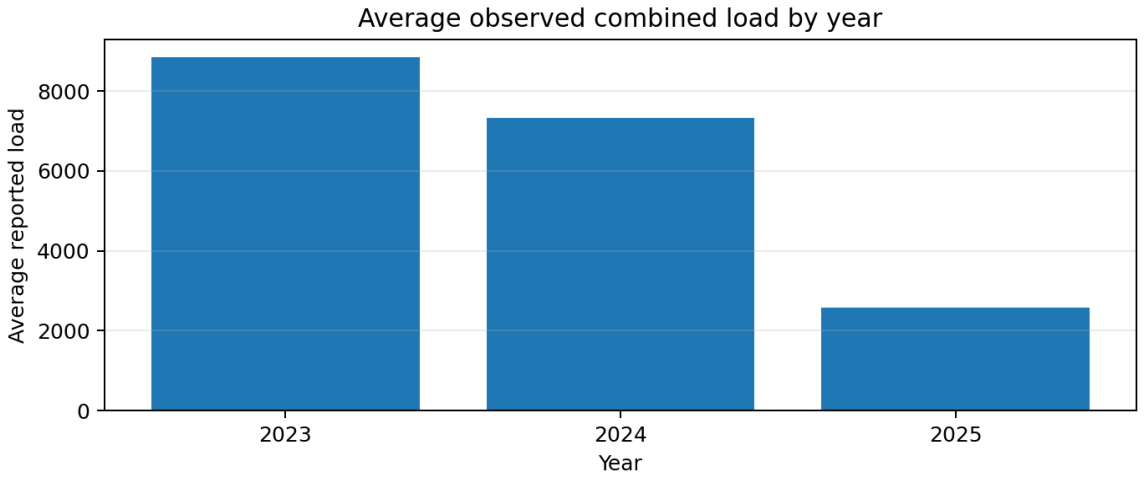


Figure 2. Average observed combined load by calendar year, calculated from the dated observations.

Year	Observed records	Average combined load	Peak combined load
2023	230	8,846.5	11,762
2024	251	7,319.9	10,994
2025	239	2,575.7	6,471

The 2025 average is lower than the 2023 and 2024 averages in the observed records. This should be interpreted alongside the source's reporting schedule and potential changes in data collection, not as proof of a particular policy or operational effect.

7. Capacity and operational interpretation

The combined load can be used as a broad situational-awareness measure, while transfer and discharge counts provide a limited view of flows. Sustained positive net intake pressure may indicate that transfers exceed discharges across a selected observation window; negative values indicate the reverse. Neither direction alone proves a capacity problem or its absence.

A defensible capacity-stress classification would require actual capacity denominators and operational context, such as licensed or staffed beds, occupied beds, temporary capacity, staff-to-child ratios, average length of stay, referral/placement wait times, and facility-level distribution. These fields are not present in the supplied dataset.

8. Recommendations

1. **Resolve reporting gaps:** confirm whether missing dates represent non-reporting, a change in publication cadence, or another source issue. Preserve missing and true zero values separately.
2. **Establish a consistent feed:** capture reporting date, source, timestamp, and revision status for each update.
3. **Add capacity denominators:** incorporate facility capacity, staffed capacity, occupancy, and temporary surge capacity.
4. **Add workforce measures:** include staffing availability and appropriate staffing ratios to assess operational pressure.
5. **Expand flow context:** add placement times, length of stay, referral queues, and facility-level transfers where appropriate.
6. **Use rolling indicators cautiously:** clearly label them as 7-/14-observation windows until daily coverage is reliable.
7. **Set review thresholds with program owners:** define escalation criteria only after validating measures and

operational context.

9. Limitations

The dataset is aggregate and does not permit person-level deduplication or cohort tracking. The combined total adds two reported snapshots and must not be interpreted as a count of unique children across the entire system. Missing reporting dates weaken continuity. Net Intake Pressure compares transfers out of CBP custody with HHS discharges and is not a complete stock-flow identity. No causal inference, forecast, or verified capacity assessment is made. The results depend on the accuracy and definitions of the supplied source fields.

10. Conclusion

The project converts dispersed aggregate counts into a structured analytical workbook with cleaned records, derived measures, temporal summaries, data-quality checks, and a decision-oriented overview. The observed records show a lower latest combined load than the first observed point and aggregate HHS discharges greater than transfers across the reporting period. These are descriptive findings only. The most important next step is to improve reporting continuity and integrate capacity, occupancy, and staffing measures so that future analysis can distinguish observed load from actual operational strain.

Appendix A. KPI definitions

KPI	Definition / interpretation
Total System Load	CBP custody snapshot + HHS care snapshot; not a deduplicated unique-person count.
Peak Total System Load	Highest observed combined snapshot in the available dated records.
Net Intake Pressure	Transfers out of CBP custody minus HHS discharges; flow-balance proxy.
Discharge Offset Ratio	Aggregate HHS discharges divided by aggregate CBP-to-HHS transfers, multiplied by 100.
Endpoint change	Percentage difference between first and latest observed combined-load values.
Rolling average	Average over 7 or 14 observed records, not necessarily consecutive calendar days.
Load volatility index	Coefficient of variation over a 7-observation window, as defined in the workbook.

Appendix B. Data quality summary

- Original source rows: 1,170.
- Fully blank rows excluded from analytical copy: 450.
- Dated observations retained: 720.
- Inclusive calendar span: 1,075 days.
- Dates without a source record: 355.
- Duplicate dates: none reported in the workbook's validation summary.
- Missing values in numeric source fields: none reported among retained dated observations.
- Negative numeric source values: none reported.

Source and reproducibility note

This paper was prepared from the project workbook UAC_System_Capacity_Care_Load_Analytics.xlsx located in the prior conversation. The workbook contains the raw-data, cleaned-data, KPI, monthly, yearly, data-quality, research-report, and executive-summary sheets. Values in this paper are reproduced from the cleaned observations and supporting analysis sheets. The original underlying CSV was not separately re-imported for this document.

End of research paper